

MATHEMATICS 2

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Mathematics 2

- 1 How many distinct real solutions does the equation $x^3 - 3x^2 + 5 = 0$ have?
- A** 0
B 1
C 2
D 3
- 2 A circle has equation $x^2 + y^2 = 50$. A tangent to the circle at a point P is parallel to the line with equation $7x + y = 1$. The point P has positive x and y coordinates.
- What are the coordinates of P ?
- A** (1, 7)
B (1, -7)
C (5, 5)
D (7, 1)
E (7, -1)
F (-7, -1)
- 3 A cubic polynomial $p(x)$ has stationary points at $x = 1$ and $x = 3$.
- What is the value of $p(2)$?
- A** -2
B -1
C 0
D 2
E 4
F Cannot be determined from the information given.

- 4 A box contains 4 red balls, 5 blue balls, and 3 green balls. All 12 balls are identical apart from their colour.

Balls are drawn at random from the box, one at a time, without replacement.

- If a red ball is drawn, it is not replaced and another ball is drawn.
- As soon as a blue ball or a green ball is drawn, the process stops.

What is the probability that at least two red balls are drawn and the process stops on a blue ball?

- A $\frac{5}{132}$
 B $\frac{1}{22}$
 C $\frac{5}{88}$
 D $\frac{5}{72}$
 E $\frac{1}{11}$
 F $\frac{5}{8}$

- 5 Find the value of the constant k for which the equation

$$2 \cos^2(x) - 3 \sin(x) = k$$

has exactly three distinct solutions in the interval $0 \leq x < 2\pi$.

- A $k = -3$
 B $k = -1$
 C $k = 2$
 D $k = \frac{25}{8}$
 E $k = 3$
 F $k = 5$

- 6** Variables x and y are related by the equation $y = Ax^n$, where A and n are constants. A graph of $\log_2 y$ plotted against $\log_2 x$ is a straight line with gradient 2 and intercept 4 on the vertical axis. Which of the following expresses y in terms of x ?
- A** $y = 4x^2$
B $y = 16x^2$
C $y = 2x^4$
D $y = 4x^4$
E $y = 2^{2x+4}$
- 7** Consider the family of quadratic curves given by
- $$y = x^2 - 2ax - 4a + 5$$
- where a is a real constant.
- What is the complete range of values of a for which the minimum point of the curve lies strictly in the second quadrant (that is, where $x < 0$ and $y > 0$)?
- A** There are no such values of a
B $a < -5$
C $-5 < a < 0$
D $-5 < a < 1$
E $0 < a < 1$
F $a < -5$ or $a > 1$
G $a < -5$ or $0 < a < 1$
- 8** The sum of the first 10 terms of a sequence is 200. Each term differs from the previous term by a constant amount. The sum of the next 10 terms is 600. What is the value of the first term?
- A** 0
B 1
C 2
D 4
E 11

- 9 A geometric series has first term a and common ratio r . All terms in the series are non-zero, and the sum to infinity of the series exists.

The sum of the first two terms is equal to $\frac{5}{9}$ of the sum to infinity. The terms in the series do not all have the same sign.

What is the third term of the series, expressed as a fraction of the sum to infinity?

- A $\frac{4}{27}$
- B $\frac{19}{27}$
- C $\frac{20}{27}$
- D $\frac{4}{9}$
- E $\frac{35}{27}$
- F $\frac{-10}{9}$

- 10 Variables x and y are related by the equation $y = kx^n$, where k and n are constants. A graph of $\log_{10} y$ against $\log_{10} x$ is a straight line passing through the points $(0, 2)$ and $(4, 0)$. Which of the following expresses y in terms of x ?

- A $y = 2 - \frac{x}{2}$
- B $y = \frac{2}{\sqrt{x}}$
- C $y = \frac{2}{x^2}$
- D $y = \frac{100}{\sqrt{x}}$
- E $y = \frac{100}{x^2}$

- 11 The function $f(x) = x^3 - 3x^2 - 9x + 30$ is defined on the restricted domain $-2 \leq x \leq 2$. What is the range of $f(x)$?

- A $3 \leq f(x) \leq 28$
- B $3 \leq f(x) \leq 35$
- C $8 \leq f(x) \leq 28$
- D $8 \leq f(x) \leq 35$
- E $-\infty < f(x) < \infty$

12 What is the coefficient of x^4 in the expansion of $(1 + 2x + 2x^2)^5$?

- A** 120
- B** 210
- C** 240
- D** 320
- E** 360

13 Consider the equation

$$\tan\left(\frac{4\pi}{x}\right) = \sqrt{3}$$

for $x > 0$.

The positive solutions are arranged in decreasing order, $x_1 > x_2 > x_3 > \dots$

What is the value of x_3 ?

- A** $\frac{6}{5}$
- B** $\frac{12}{13}$
- C** $\frac{12}{7}$
- D** $\frac{24}{13}$
- E** 3
- F** 12

14 A sector of a circle has a fixed perimeter of 60.
What is the radius of the sector that maximises its area?

- A** $30/\pi$
- B** $60/(\pi + 2)$
- C** 15
- D** 20
- E** 30

- 15** The positive numbers α and β satisfy the relation $\alpha = 3\beta$. A point P on the y -axis is a distance of 5 from the point $(\alpha, 0)$ and a distance of 3 from the point $(\beta, 0)$.

What is the value of the product $\alpha\beta$?

- A** 2
- B** 6
- C** 7
- D** 16
- E** 18
- F** 24

- 16** A solid right circular cylinder has base radius r and height h , as shown in the diagram. The base radius r is decreased by 20%, while the volume of the cylinder is increased by 28%. By what percentage does the height h increase?

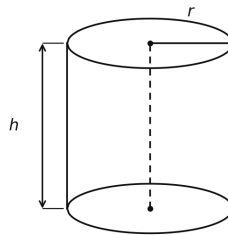


Diagram not to scale

- A** 48%
- B** 50%
- C** 60%
- D** 68%
- E** 100%
- F** 200%

- 17 What is the value of the sum

$$S = \sum_{n=2}^{49} \log_{10} \left(1 - \frac{1}{n^2} \right)$$

- A $\log_{10} \left(\frac{25}{49} \right)$
 - B $\log_{10} \left(\frac{1}{2} \right)$
 - C $\log_{10} \left(\frac{50}{49} \right)$
 - D $\log_{10} \left(\frac{49}{50} \right)$
 - E $\log_{10} \left(\frac{24}{25} \right)$
- 18 Find the area of the finite region in the first quadrant bounded by the curve $y = 2x^2$, the line $y = 32$, and the y -axis.
- A $\frac{128}{3}$
 - B 64
 - C $\frac{256}{3}$
 - D 128
 - E $\frac{320}{3}$
 - F $\frac{512}{3}$

- 19** $ABCD$ is a trapezium with parallel sides AB and CD . The diagonals AC and BD intersect at point X , as shown in the diagram.

$AX = 1$ cm, $AB = x$ cm, $CX = (x + 2)$ cm, and $CD = (4x + 1)$ cm.

What is the length, in cm, of CD ?

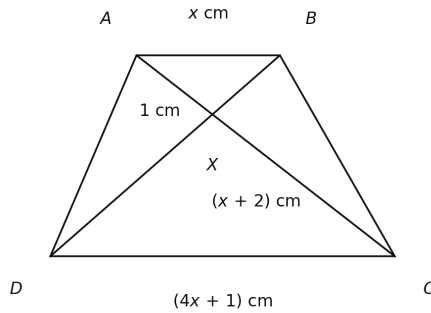


Diagram not to scale

- A** $3 + 2\sqrt{5}$
 - B** $4 + \sqrt{2}$
 - C** $4 + 4\sqrt{2}$
 - D** $5 + 4\sqrt{2}$
 - E** $5 + 4\sqrt{3}$
 - F** $5 + 8\sqrt{2}$
- 20** The polynomial $(x + y)^{37}$ is expanded in ascending powers of x .
How many of the terms in this expansion have a coefficient that is divisible by 37?
- A** 35
 - B** 36
 - C** 37
 - D** 38
 - E** 0

- 21** Which one of the following statements describes a mathematically impossible scenario for a polynomial $p(x)$ with real coefficients?
- A** A quadratic polynomial $p(x)$ has two distinct positive real roots and its derivative at $x = 0$ is positive.
 - B** A cubic polynomial $p(x)$ has a local maximum at $x = 1$, a local minimum at $x = 4$, and $p(1) < p(4)$.
 - C** A cubic polynomial $p(x)$ has exactly one real root, and its derivative $p'(x)$ has two distinct real roots.
 - D** A cubic polynomial $p(x)$ has no stationary points and passes through the origin.
 - E** A quartic polynomial $p(x)$ has exactly one stationary point.
 - F** A cubic polynomial $p(x)$ has a local minimum at $x = 1$ and a local maximum at $x = 4$.

- 22** A box contains 12 electronic components, of which 4 are defective and 8 are working. Components are drawn at random from the box, one at a time without replacement, and tested sequentially.

Testing stops immediately once a second defective component has been identified.

What is the probability that fewer working components than defective components have been tested when the process stops?

- A** $\frac{1}{11}$
- B** $\frac{8}{55}$
- C** $\frac{9}{55}$
- D** $\frac{13}{55}$
- E** $\frac{17}{55}$
- F** $\frac{23}{55}$

- 23** A function $f(x)$ is defined for all real x , is differentiable at $x = 0$, and satisfies the equation:

$$f(x) = 2f\left(\frac{x}{3}\right) + x^2 + x$$

What is the value of $f(7)$?

- A** 21
- B** 56
- C** 63
- D** 84
- E** 168

- 24** The function $f(x)$ is defined by

$$f(x) = \frac{5^x}{5^x + \sqrt{5}}$$

What is the value of the following sum?

$$S = \sum_{r=0}^{100} f\left(\frac{r}{100}\right)$$

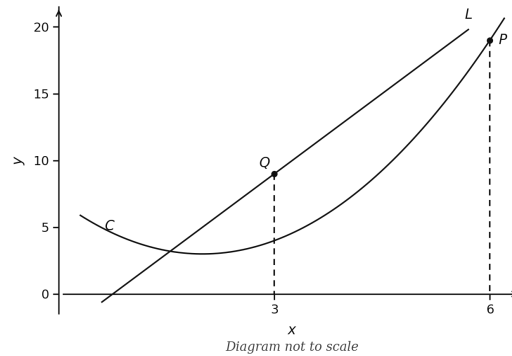
- A** 49.5
- B** 50
- C** 50.5
- D** 51
- E** 100
- F** 101

- 25** The diagram shows part of the curve C with equation $y = (x - 2)^2 + 3$ and part of the straight line L with equation $y = 4x + c$, where c is a constant.

The point P lies on C at $x = 6$, and the point Q lies on L at $x = 3$.

The y -coordinate of P is equal to twice the y -coordinate of Q plus 1.

What is the y -coordinate of the point on L with x -coordinate 5?



- A** -3
B 9
C 17
D 17.5
E 23
F 25
G 47
- 26** A right circular cone has base radius r and perpendicular height h . The volume of the cone is V and its total surface area is A . These quantities are related by the equation:

$$\ln(9V) - \ln(A) = \ln(r)$$

Which of the following expresses h in terms of r ?

- A** $h = \frac{3}{4}r$
B $h = \frac{1}{2}r$
C $h = \frac{3}{8}r$
D $h = \frac{4}{3}r$
E $h = \frac{9}{40}r$
F $h = \frac{\sqrt{2}}{4}r$

27 Let P and Q be positive real numbers.

Q is the value obtained after P is first increased by 100%, then the result is decreased by 75%, and finally that result is increased by 100%.

What is the value of $\log_4(Q) - \log_4(P)$?

- A** -1
- B** $-\frac{1}{2}$
- C** 0
- D** $\frac{1}{2}$
- E** 1